Published online 17 May 2023 *Nucleic Acids Research, 2023, Vol. 51, Web Server issue* ***W601–W606***
https://doi.org/10.1093/nar/gkad406

WebQUAST: online evaluation of genome assemblies

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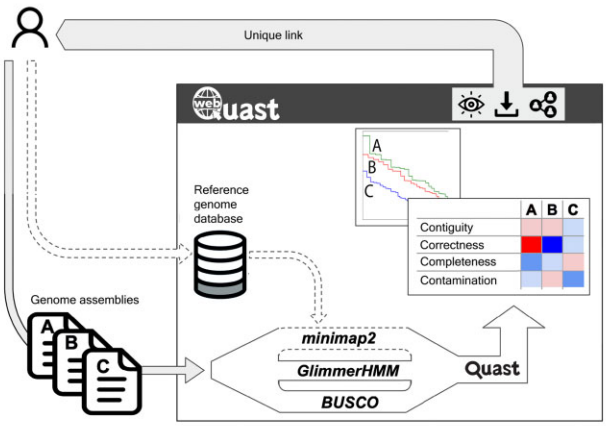
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Received March 31, 2023; Revised April 25, 2023; Editorial Decision May 03, 2023; Accepted May 03, 2023

ABSTRACT

Selecting proper genome assembly is key for downstream analysis in genomics studies. However, the availability of many genome assembly tools and the huge variety of their running parameters challenge this task. The existing online evaluation tools are limited to specific taxa or provide just a one-sided view on the assembly quality. We present WebQUAST, a web server for multifaceted quality assessment and comparison of genome assemblies based on the state-of-the-art QUAST tool. The server is freely available at https://www.ccb.uni-saarland.de/quast/. WebQUAST can handle an unlimited number of genome assemblies and evaluate them against a user-provided or pre-loaded reference genome or in a completely reference-free fashion. We demonstrate key WebQUAST features in three common evaluation scenarios: assembly of an unknown species, a model organism, and a close variant of it.

GRAPHICAL ABSTRACT



INTRODUCTION

Despite the ongoing long-read sequencing revolution, it is still impossible to read entire chromosomes for most species in a single run (1). Researchers use the so-called genome assembly software that combines the sequencing reads into longer genome fragments commonly referred to as contigs. Dozens of genome assemblers exist nowadays (2). These tools rely on different heuristics that greatly vary their output. Moreover, even different settings of the same tool may result in substantially diverging assemblies. The quality assessment and comparison of multiple genome assemblies are of utmost importance since the assembly choice greatly affects the downstream analysis (3).

The existing assembly evaluation tools comprise two major categories. The reference-based tools, such as GAGE (4), use gold-standard reference genomes to evaluate assemblies on model datasets. The reference-free methods either rely on read mapping back to assemblies to check their consistency with the input data and detect assembly er-

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rors, such as REAPR (5) and Inspector (6), or look for conservative genes to estimate the assembly completeness, such as BUSCO (7,8) and CEGMA (9). Previously, we developed QUAST, an ensemble method that incorporated the best software from both categories, enhanced them with in-house quality metrics and plots, and became the state-of-the-art quality assessment tool for genome assemblies (10,11). However, QUAST intrinsically inherited the limitations of the embedded tools which are available only for a few platforms (usually Linux) and have a command-line interface making them hardly suitable for researchers with a limited computational background.

Here, we present WebQUAST, a web server complementing QUAST with a user-friendly graphical interface and providing its functionality on any platform. In contrast to a few existing genome assembly evaluation web tools, WebQUAST is not restricted to specific taxa as gEVAL (12) and GenomeQC (13), performs versatile assembly evaluation rather than only completeness estimation as gVolante (14), and supports an unlimited number of assemblies on input. The WebQUAST evaluation reports can be browsed online, downloaded locally, and shared privately with colleagues. We show WebQUAST performance using a sample dataset of four *E. coli* assemblies.

MATERIALS AND METHODS

Web server overview

Workflow. A user uploads genome assemblies in the FASTA format (gzipped files are supported), configures the evaluation parameters, such as the minimal contig length cut-off and the organism type (eukaryote or prokaryote), and optionally selects a reference genome. The user might choose it from the list of pre-loaded genomes or upload a custom FASTA file that will be stored privately and can be reused later. Once the user clicks on the Evaluate button, WebQUAST transfers the input data to the QUAST processing engine.

If a reference genome is provided, the assemblies are aligned against it using minimap2 (15). If the BUSCO checkbox is selected, the assemblies are screened for single-copy orthologues from the corresponding BUSCO database (8). If the gene finding is requested, the assemblies are processed with the GlimmerHMM gene prediction software (16). QUAST combines the outputs of all employed modules to compute numeric quality metrics, create assessment plots and Icarus viewers (17), and generate a single evaluation report. WebQUAST assigns the report a unique web link and renders it for the user. The link enables browsing the results online and sharing them. The user can download the full standalone report to store it permanently. The standalone report also provides additional insights into the analysis, such as the running commands of the embedded tools or the list of identified misassemblies in the GFF format.

Software implementation. The server is built on top of the Python web framework Django. MySQL instance is used to record users, sessions, and analysis requests. To support long-running analysis, the requests are processed and added into an asynchronous task queue Celery. A queued job represents a simple script that calls the command-line QUAST tool, which allows us to keep the main codebase agnostic to the web implementation. The front-end component is based on the jQuery framework.

Sample data preparation

To demonstrate WebQUAST performance, we generated sample assemblies of a well-studied short-read *Escherichia coli* K-12 MG1655 dataset (SRA accession: ERR008613). The choice of a genome assembler might be influenced by many factors and one popular, yet often suboptimal, strategy is to choose among the most-cited methods (18). We mimicked this behavior by collecting information on short-read genome assemblers (Table 1) and selecting the five most-cited tools. We further excluded SOAPdenovo (19) since the authors discontinued it and recommended using MEGAHIT (20), which was already shortlisted.

Some of the selected assemblers do not include a read error correction module, so we cleaned the raw sequencing data beforehand to make the comparison fair. We checked the reads with FastQC and trimmed low-quality ends with Trimmomatic (37). All assemblers but ABySS were run with default parameters or based on the recommendations in the documentation wherever available. We used the GAGE-B recipe (38) for ABySS since its default assembly was of very poor quality. All tools were installed via Bioconda (39), the installation and running commands are in the Supplementary Material.

RESULTS

Here we illustrate three typical WebQUAST usage scenarios. In each case, we evaluated the same four assemblies of the *E. coli* K-12 MG1655 dataset but selected the reference genome differently. We assumed the reference was unknown in Case 1, exactly matched the dataset in Case 2, and was closely related to the dataset in Case 3.

Use Case 1: reference-free evaluation

When a reference genome is unavailable, WebQUAST computes 30 quality metrics and draws three assessment plots that mainly address the contiguity and completeness of the provided assemblies (Figure 1A, Supplementary Figure S1). The heatmaps help to detect the best-performing tools in each category.

Figure 1A shows that there is no single winner in all metrics. Compared to three other methods, ABySS produced the largest (4.8 Mb versus 4.6 Mb) but also the most fragmented assembly (176 contigs versus 90–95 for Velvet, SPAdes and MEGAHIT). SPAdes assembled larger contigs on average (the best N50, N90 and auN, the area under the Nx curve, values with Velvet and MEGAHIT being close runner-ups) and has the largest contig overall (285 versus 265, 248 and 236 kb for Velvet, ABySS and MEGAHIT). The MEGAHIT assembly does not contain uncalled bases ('N') while Velvet has the most of them (94 per 100 kb). All four assemblies are equally complete in terms of fully assembled representative bacterial single-copy orthologs (98.7%

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Table 1. The most-cited short-read genome assemblers

Assembler	Latest release version (year)	Num citations total	Num citations yearly	Key publications with years
SPAdes	3.15.5 (2022)	18833	1847	2020 (21), 2012 (22)
Velvet	1.2.10 (2014)	10633	709	2008 (23)
SOAPdenovo	242 (2018)	7410	630	2012 (19), 2010 (24)
MEGAHIT	1.2.9 (2019)	4519	581	2016 (20), 2015 (25)
ABySS	2.3.5 (2022)	4445	366	2017 (26), 2009 (27)
IDBA	1.1.3 (2016)	2979	266	2012 (28), 2010 (29)
ALLPATHS	52488 (2016)	2868	210	2011 (30), 2008 (31)
MaSuRCA	4.1.0 (2023)	1434	164	2017 (32), 2013 (33)
Ray	2.3.1 (2014)	1232	103	2012 (34), 2010 (35)
SGA	0.10.15 (2016)	909	83	2012 (36)

Version numbers and dates of the latest release were determined from the GitHub repositories of the tools. *Num citations* stands for the number of citations according to Google.Scholar as of 28.03.2022, *yearly* average is the total number of citations divided by the sum of full years past since the publications. At most two key publications per tool are included; if there were more than two publications, we relied on the citation recommendations on the tool webpage (usually the first and the last publication).

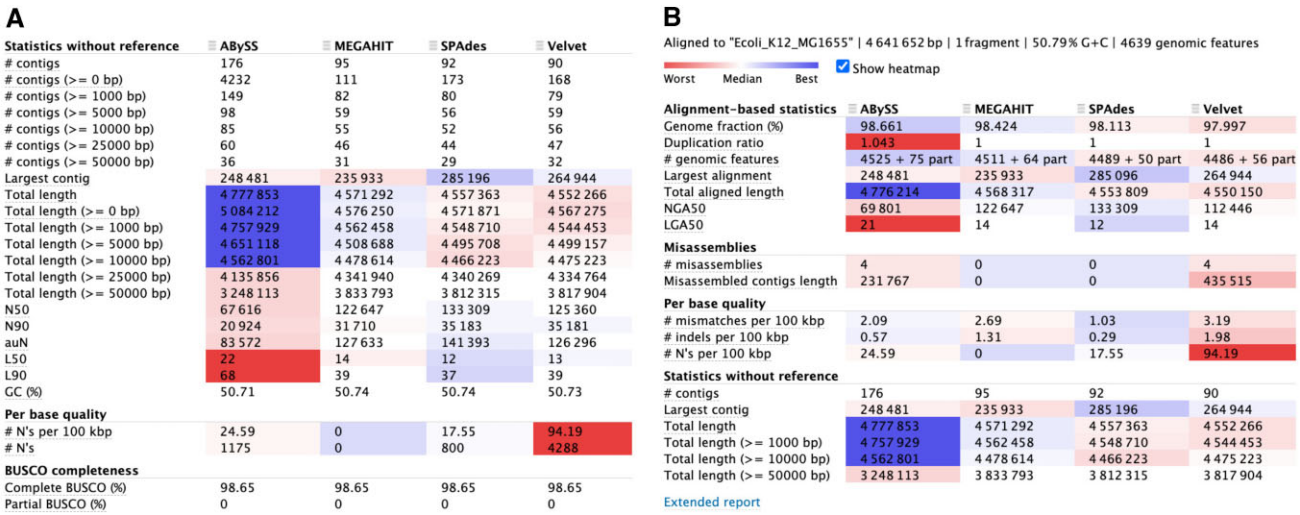


Figure 1. WebQUAST text reports for *E. coli* assemblies in the (**A**) reference-free and (**B**) reference-based evaluation mode. Unless otherwise noted, all statistics are based on contigs of size ≥ 500 bp (the default cut-off). Heatmap highlights the best value in each row which could be the largest or the smallest number depending on the quality metric. Heatmap is not used for *# contigs* and *GC (%)* due to the ambiguity of these metrics trends.

of the BUSCO genes). The average G + C content of all assemblies (50.7%) perfectly matches the expected range for *E. coli* (50.4–50.8% (40)) indicating the likely absence of contaminants in the dataset. This hypothesis is further supported by the GC plot (Supplementary Figure S1D), though we cannot exclude a presence of an organism with similar G + C content.

Use Case 2: reference-based evaluation

A reference genome enables accurate and versatile evaluation by WebQUAST in all four quality categories: contiguity, correctness, completeness, and contamination. In this mode, the tool reports >60 quality metrics accompanied by eight assessment plots and two Icarus viewers (Figure 1B, Figure 2A, Supplementary Figures S2–S4). By default, WebQUAST displays only 18 key metrics and hides the rest behind the Extended report button (Figure 1B).

As in Use Case 1, there is no undisputed best assembly in Figure 1B. However, we can now investigate some quality categories in more detail. The increased Duplication ratio for ABySS (1.04 versus 1.00 for the rest assemblers) indicates that this method assembled many genomic regions more than once. Still, ABySS assembled the highest percentage of the genome (98.7 versus 98.0-98.4% for Velvet, SPAdes and MEGAHIT) but its leadership is not as evident as it appeared when we compared the total assembly lengths. SPAdes and ABySS have the best per-base quality with SPAdes being twice better as the runner-up (1.0 vs 2.1 mismatches and 0.3 versus 0.6 indels per 100 kb). MEGAHIT and SPAdes made no large assembly errors, while Velvet and ABySS have four misassemblies each. Though, the largest contigs in all four assemblies are error-free since their lengths exactly match the largest alignments. The Icarus viewer can be used for deep inspection of the misassembly locations (Figure 2, Supplementary Figure S4).

Use case 3: evaluation based on a close reference

The true reference genome is rarely known in real studies but a close reference could often be available. Here we used

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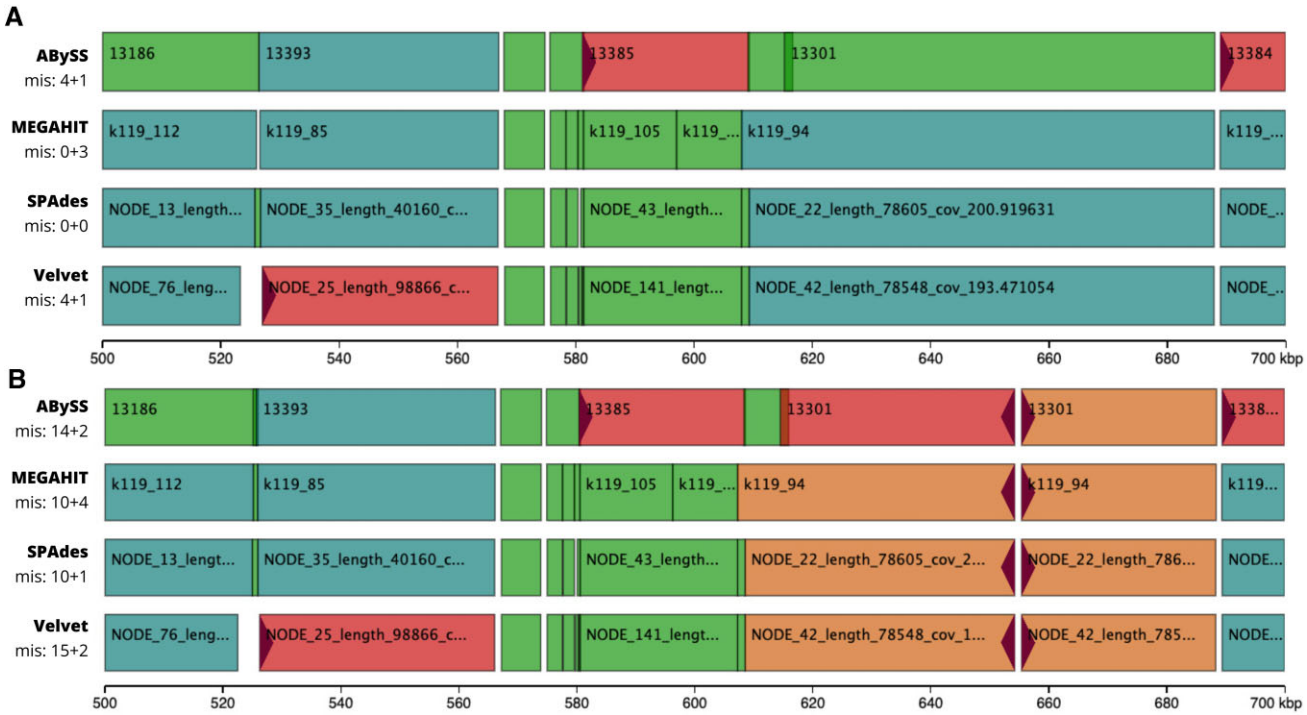


Figure 2. Icarus viewers for *E. coli* assemblies aligned against (**A**) the reference genome matching the dataset and (**B**) a close reference. The reference regions between 0.5 Mb and 0.7 Mb are shown. *mis: X + Y* stands for the total number of extensive (*X*) and local (*Y*) misassemblies per assembly. Correctly assembled contigs are colored green and aquamarine (if longer than 10 kb and similar in at least three assemblies), and fragments of misassembled contigs are colored pink and orange (if similar in at least three assemblies). Red triangles designate the sides of alignment breakpoints for misassembled contigs. Contig names are shown for contigs of sufficient size.

W3110, another *E. coli* K-12 substrain, as an example of a close reference (Figure 2B, Supplementary Figures S5–S7). Naturally, the absolute values of many alignment-based metrics, such as lengths of misassembled and unaligned contigs, substantially deteriorated due to the actual differences between the sequenced organism and the provided reference genome. However, they are still useful for determining the best assembly among available options.

Figure 2 highlights the substantially increased number of misassemblies compared to the evaluation based on the true reference genome (49 versus 8 extensive misassemblies in total). However, a closer look at the misassembly locations, suggests that almost all of them are the same in all assemblies which likely means they are true structural variations rather than assembly errors and can be ignored for evaluation purposes (Figure 2B and Supplementary Figure S7). Though, we cannot exclude the possibility that several assemblers made the same error in a complex genomic region, especially if we compare tools inspired by the same computational approach such as the de Bruijn graph-based assembly (41).

CONCLUSION

Selecting the best – or, more precisely, the most suitable – genome assembly is crucial for downstream analysis. While many post-processing steps, such as structural and functional annotation (42) or genome mining (43), have been available online for years, the assembly validation step is still mainly done with the Linux-based command-line tools. Here, we presented WebQUAST, a web server for genome assembly evaluation, that greatly facilitates this task for users with any operating system and computational background and helps them to make an informed choice. Since our tool is suitable for any organism and sequencing technology, we expect it would benefit the broad genomics community. Furthermore, WebQUAST is already incorporated in several bioinformatics massive online open courses (MOOCs), so we hope it would also help to educate the future generation of researchers.

DATA AVAILABILITY

WebQUAST is freely available at https://www.ccb.uni-saarland.de/quast/. The source code for the server is at https://github.com/ablab/quast-website and for the core QUAST tool is at https://github.com/ablab/quast. The sequencing data for *E. coli* K-12 MG1655 dataset is available from the National Center for Biotechnology Information (NCBI) Sequence Read Archive under accession number ERR008613. The *E. coli* strain K-12 reference genomes and gene annotations are available from NCBI under accession numbers NC_000913.3 and AP009048.1 for substrains MG1655 and W3110, respectively. The ABySS, MEGAHIT, SPAdes, and Velvet assemblies generated in this study and their interactive evaluation reports are available from the WebQUAST front page and in Zenodo at https://doi.org/10.5281/zenodo.7863703.

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SUPPLEMENTARY DATA

Supplementary Data are available at NAR Online.

ACKNOWLEDGEMENTS

We thank St. Petersburg Academic University for hosting WebQUAST in 2012–2020 and the Computer Center of Research Park of St. Petersburg State University for hosting it in 2020–2023.

FUNDING

Funding for open access charge: Helmholtz Centre for Infection Research.
Conflict of interest statement. None declared.

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